

Chapter 5 — Power Supplies, Breakdown Diodes, and Voltage References

Electronics Device and Circuit (ENEX 151 / EX 151) · Tribhuvan University, IOE · BE (BEI, BCT) · Year I / Part II

Exam-oriented notes with embedded Past-Year Questions (2081 Ashwin R · 2082 Baishakh B · 2082 Bhadra R · 2083 Baishakh B)

References: Boylestad/Floyd (regulator and protection circuits, LM317 design pattern with $R_1 = 240\ \Omega$), Millman-Halkias (stability analysis), Sedra-Smith (zener model, bandgap reference)

Figures: 7 diagrams rendered locally — filter ripple and the LM317 design chart computed from the actual equations with the PYQ values (fetch report: 0/7 from web)

How to Use These Notes / Chapter Overview

From wall AC to clean DC: **transformer** → **rectifier** → **filter** gives an *unregulated* supply that sags and ripples; a **regulator** fixes it. The chapter climbs a ladder — zener clamp → transistor series/shunt regulator → feedback-improved regulator with protection → IC regulator (LM317). The exam samples this chapter for **8-9 marks every year with the same two questions**: an LM317 range design and one sub-topic of the series-regulator family. This is the cheapest guaranteed 8 marks on the paper.

PYQ Frequency Analysis — Chapter 5 = 8-9 marks every single year

Topic	2081 Ashwin	2082 Baishakh	2082 Bhadra	2083 Baishakh	Verdict
LM317 variable-regulator design	✓ 10-20 V [4]	✓ 5-15 V [4]	✓ 10-25 V [4]	✓ 3-15 V [4]	★ 4/4 GUARANTEED — only the range changes
Series transistor regulator (working / protection / stability)	✓ improved + stability expr. [2+2]	✓ current limiting [4]	✓ short-circuit / overload protection [5]	✓ stability-factor derivation [4]	★ 4/4 GUARANTEED — sub-angle rotates
Unregulated supply / zener / references	-	-	-	-	foundation only — never standalone

Strategic conclusions

- ★ The **LM317 design is one formula evaluated twice**. Four papers, four ranges, identical method — a rehearsed template answers it in four minutes. All four PYQ designs are computed in §5.6 and plotted on one chart.
- ★ The series-regulator question **rotates its sub-angle**: stability factor (2083) → overload protection (2082 Bhadra) → current limiting (2082 Baishakh) → feedback improvement + stability (2081). Prepare the family as ONE circuit that grows: follower → + error amplifier → + sense resistor. Then every rotation is a subset.
- §5.1-5.3 have never been asked standalone, but §5.3's bandgap IS the LM317's internal 1.25 V — citing that link upgrades the design answer.

5.1 The Unregulated Power Supply

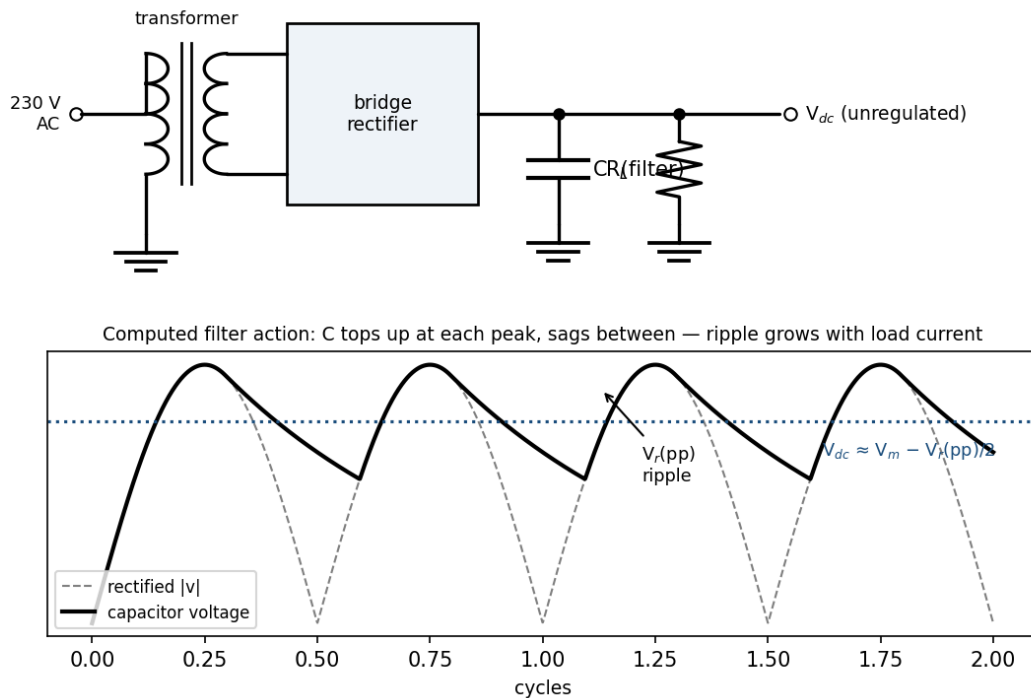


Fig. 1 — The unregulated chain, and the computed filter waveform: the capacitor charges to each rectified peak and discharges into the load between peaks. [circuit rendered locally; waveform computed]

- **Chain:** transformer (steps the mains to the required amplitude and provides isolation) → rectifier (half-wave, full-wave centre-tap, or bridge) → capacitor filter → load.
- **Key quantities:** $V_{dc} \approx V_m - V_r(pp)/2$; **ripple factor** $r = V_r(rms)/V_{dc}$; for full-wave + C filter, $V_r(rms) = I_{dc}/(4\sqrt{3} \cdot f \cdot C)$ — Boylestad's working form: $2.4 \cdot I_{dc}/C$ (I in mA, C in μF , %r referenced to V_{dc}). Ripple grows with load current.

Voltage regulation: $\%VR = (V_{NL} - V_{FL})/V_{FL} \times 100\%$ — unregulated supplies have poor %VR and ride the mains; hence the rest of this chapter

5.2 Zener-Regulated Power Supply

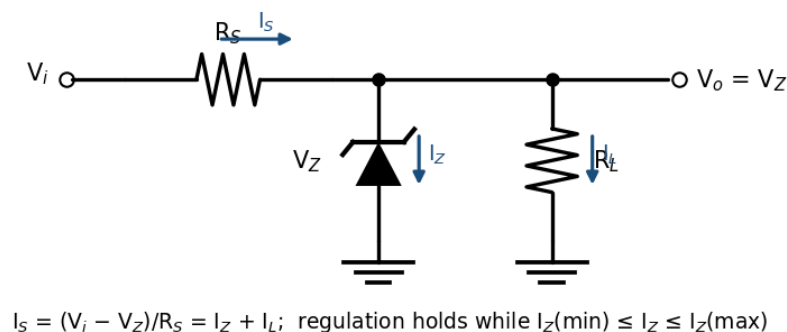


Fig. 2 — The zener shunt clamp: R_S drops the excess voltage; the zener absorbs whatever current the load does not take. [rendered locally]

- **Operating rule:** the zener holds $V_o = V_Z$ as long as it stays in breakdown: $I_{Z(min)} \leq I_Z \leq I_{Z(max)}$, with $I_S = (V_i - V_Z)/R_S = I_Z + I_L$. Load takes more → zener takes less, and vice versa.

- **Design check-list:** pick R_S small enough that $I_Z \geq I_{Z(\min)}$ at worst case ($V_{i(\min)}$, $I_{L(\max)}$); then verify $P_Z = V_Z \cdot I_{Z(\max)}$ at the other extreme ($V_{i(\max)}$, no load).
- **Limits — state them, they motivate §5.4:** load current capped by zener power; output stiffness only as good as the zener's dynamic resistance ($\Delta V_o = r_z \cdot \Delta I_Z$); R_S burns power continuously.

5.3 Breakdown Diodes, Bandgap Reference, Constant-Current Diodes

- ● **LOW Breakdown physics:** true **zener** breakdown (quantum tunnelling) dominates below ≈ 5 V and has a **negative** temperature coefficient; **avalanche** breakdown (impact ionisation) dominates above ≈ 6 V with a **positive** TC. Around 5.6–6.2 V the two mix to nearly zero TC — the best reference diodes. Circuit model: ideal V_Z in series with small r_z .
- **Bandgap reference:** add a PTAT voltage ($\Delta V_{BE} = V_T \ln n$ between two BJTs, +TC) to a plain V_{BE} (-2 mV/°C) with weights chosen so the sum sits at silicon's bandgap $\approx$ **1.205–1.25 V** — the TCs cancel. **This is the internal 1.25 V reference of the LM317** (§5.6): drop that sentence in the design answer for polish.
- **Constant-current diode:** a JFET with gate shorted to source — pins the current at I_{DSS} once $V > |V_p|$; the current-domain dual of the zener (ties back to Chapter 2).

5.4 Transistor Series and Shunt Regulators ★ REPEATED PYQ TOPIC — the base circuit of FOUR PYQs

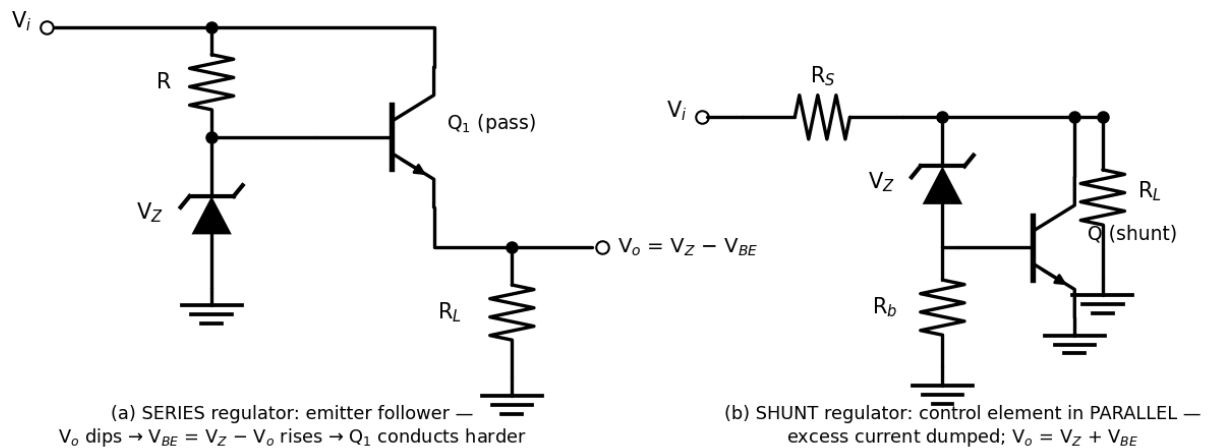


Fig. 3 — (a) The series (emitter-follower) regulator — the skeleton every protection/feedback/stability PYQ builds on. (b) The shunt regulator for contrast. [rendered locally]

Series regulator — operation (the narration every answer opens with)

- The pass transistor Q_1 stands **in series** between input and load, working as an **emitter follower** off the zener: $V_o = V_Z - V_{BE}$.
- Self-correction:** if V_o dips (load current up, or V_i down), then $V_{BE} = V_Z - V_o$ rises, Q_1 conducts harder, and V_o is pulled back up — continuously, with no clock and no drama. The symmetric argument handles V_o rising.
- Current leverage:** the zener now supplies only *base* current — the regulator delivers $\beta \times$ the zener's capability. That, plus efficiency at light load, is why the series type is the mainstream choice.
- Shunt contrast (1-2 marks when asked):** control element in *parallel* with the load, $V_o = V_Z + V_{BE}$, excess current deliberately dumped. Inherently short-circuit-proof but always burning power.

★ Stability factor of the discrete-transistor series regulator — PYQ 2083 Baishakh Q9 [4]

- Definition [1]:** $S_V = \Delta V_o / \Delta V_i$ at constant load — the fraction of input (line/ripple) variation that leaks through to the output. Good regulator $\Rightarrow S_V \ll 1$.
- Derivation [2]:** $V_o = V_Z - V_{BE}$, and for small changes $\Delta V_{BE} \approx 0$, so $\Delta V_o = \Delta V_Z$. Model the zener as $V_{Z0} + r_z$. In Fig. 3(a) the zener is fed **from V_i through R** , so an input wiggle Δv_i simply divides across the R - r_z chain: $\Delta v_z = \Delta v_i \cdot r_z / (R + r_z)$. Therefore:

$$S_V = \Delta V_o / \Delta V_i = r_z / (R + r_z) \blacksquare$$

- Interpretation [1]:** small r_z (stiff zener) and large $R \Rightarrow$ strong line rejection. Numbers: $r_z = 10 \Omega$, $R = 1 \text{ k}\Omega \Rightarrow S_V \approx 0.0099$ — about 1% of input ripple reaches the load. Improvements to name: feed R from the *output* (pre-regulation), or close a feedback loop (§5.5), which divides S_V further by the loop gain.
- ⚠ Mistakes: writing $V_o = V_Z + V_{BE}$ for the series type (that is the shunt) · deriving with the zener fed from V_o while drawing it fed from V_i — state which topology you analyse · forgetting to say $\Delta V_{BE} \approx 0$.

5.5 Improving Regulator Performance — Feedback and Protection ★ REPEATED PYQ TOPIC

A. Feedback (error-amplifier) series regulator — PYQ 2081 Ashwin Q13 [2+2]

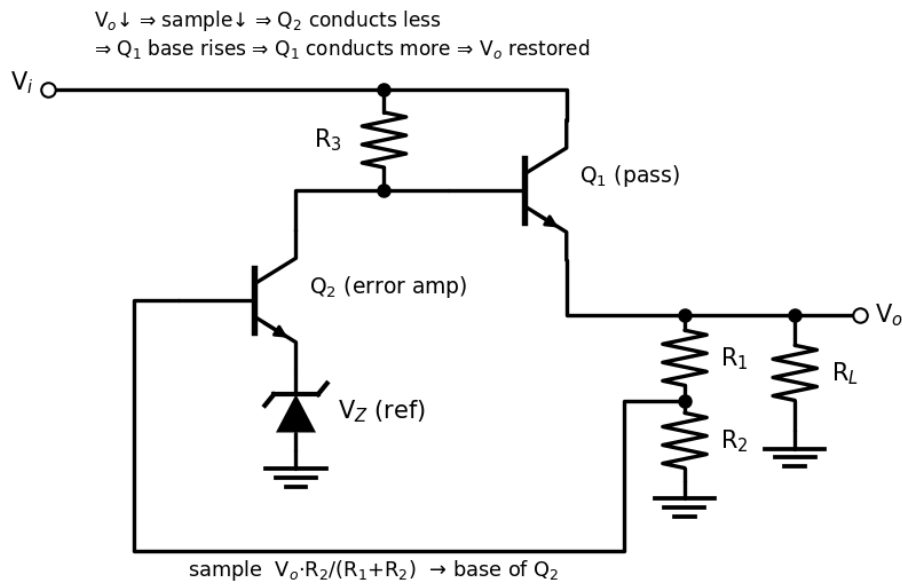


Fig. 4 — Feedback series regulator: divider R_1 - R_2 SAMPLES V_o ; error amplifier Q_2 COMPARES the sample against the zener reference and drives the pass transistor's base through R_3 . [rendered locally]

- **Operation [2]:** $V_o \downarrow \Rightarrow \text{sample } V_o R_2 / (R_1 + R_2) \downarrow \Rightarrow V_{BE2} \downarrow \Rightarrow Q_2 \text{ conducts less} \Rightarrow \text{less drop across } R_3 \Rightarrow Q_1 \text{'s base voltage } \uparrow \Rightarrow Q_1 \text{ conducts more} \Rightarrow V_o \text{ restored (and vice-versa). The loop settles when the sample equals the reference stack } V_Z + V_{BE2}$:

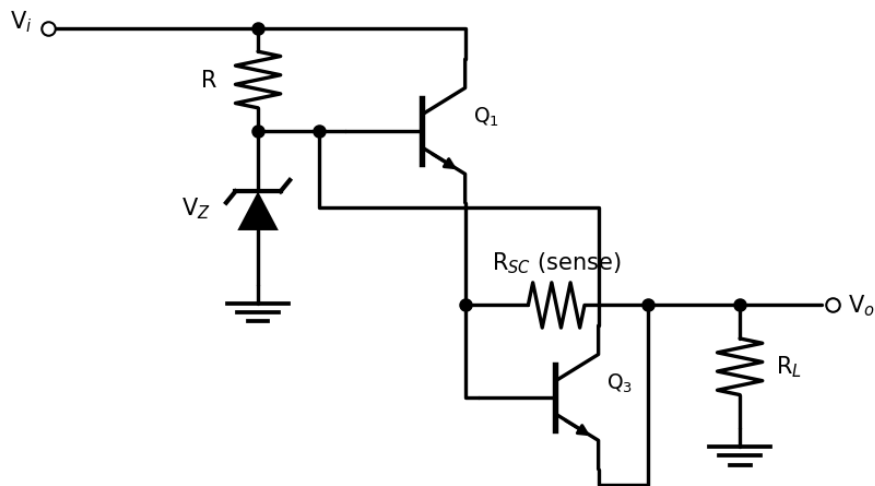
$V_o = (V_Z + V_{BE2}) \cdot (R_1 + R_2) / R_2$ — now ADJUSTABLE (make part of the divider a pot), and set by resistor ratios, not by the pass device

- **Voltage-stability expression [2]:** negative feedback divides the open-loop feed-through by the loop gain:

$S_V(\text{closed}) \approx S_V(\text{open}) / (1 + A\beta_{fb}) = [r_z / (R + r_z)] \cdot 1 / (1 + A\beta_{fb})$ · likewise
 $R_{out}(\text{closed}) = R_{out} / (1 + A\beta_{fb})$

- 'Improved performance' therefore means better **line** regulation AND better **load** regulation, both by the same loop-gain factor — say both for the full 2 marks. (With an op-amp as the error amplifier, A is huge and $V_o \rightarrow V_Z(R_1 + R_2) / R_2$.)

B. Short-circuit / overload (current-limiting) protection — PYQs 2082 Bhadra Q9 [5], 2082 Baishakh Q12(a) [4]



$V_{BE3} = I_L \cdot R_{SC}$: below 0.7 V $\Rightarrow$ Q_3 off (no effect);
at ≈ 0.7 V $\Rightarrow$ Q_3 on, steals Q_1 's base drive $\Rightarrow$ I_L clamped at $\approx 0.7/R_{SC}$

Fig. 5 — Current limiting: sense resistor R_{SC} in the output path develops $I_L \cdot R_{SC}$ across Q_3 's base-emitter; at ≈ 0.7 V, Q_3 turns on and diverts Q_1 's base drive. [rendered locally]

- **Normal load:** drop $I_L R_{SC} < 0.7$ V $\Rightarrow$ Q_3 OFF — the protection circuit is invisible and regulation proceeds normally.
- **Overload / short:** the drop reaches ≈ 0.7 V $\Rightarrow$ Q_3 turns ON and **steals Q_1 's base drive** (its collector pulls the base node down). Q_1 can then pass no more than:

$$I_{L(\max)} \approx V_{BE(on)}/R_{SC} \approx 0.7/R_{SC} \quad (\text{e.g. } R_{SC} = 1 \, \Omega \Rightarrow \text{limit} \approx 0.7 \, \text{A})$$

- **Behaviour into the fault:** the regulator degenerates into a *constant-current source*; V_o collapses but Q_1 and the load survive, and normal operation resumes when the fault clears. **Refinement to name:** foldback limiting — reduces the limited current as V_o falls, cutting short-circuit dissipation in Q_1 .
- $\triangle$ Mistakes: placing R_{SC} where it does not carry the load current · Q_3 's b–e not across R_{SC} · claiming the output voltage is protected (it is the *current* that is clamped; the voltage collapses by design).

5.6 IC Voltage Regulators — the LM317 ★ REPEATED PYQ TOPIC (designed EVERY year)

Fixed three-terminal families first, one line each: **78xx** = fixed positive (7805 → +5 V), **79xx** = fixed negative; internally a reference + error amplifier + pass device + protection, i.e. \$5.5 in a package. The star is the adjustable **LM317**.

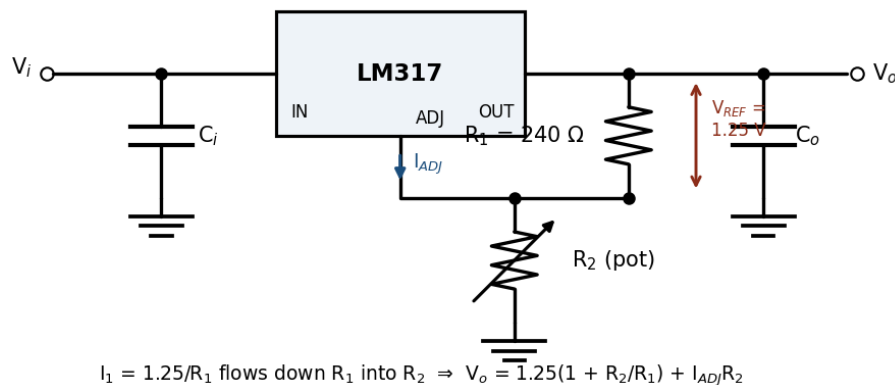


Fig. 6 — The LM317 design circuit: the IC holds its internal bandgap reference $V_{REF} = 1.25$ V between OUT and ADJ, so R_1 carries a fixed $I_1 = 1.25/R_1$ which flows on into R_2 . C_i/C_o are decoupling. [rendered locally]

Where the formula comes from (two lines — write them before substituting)

- 1. The IC regulates $V_{REF} = 1.25$ V between OUT and ADJ $\Rightarrow$ a constant $I_1 = V_{REF}/R_1$ flows down R_1 .
- 2. That current (plus the tiny $I_{ADJ} \approx 50\text{--}100$ μA leaving the ADJ pin) flows through R_2 , stacking R_2 's drop on top of V_{REF} :

$$V_o = V_{REF}(1 + R_2/R_1) + I_{ADJ}R_2 \approx 1.25(1 + R_2/R_1) \text{ V} \Rightarrow \text{design inversion: } R_2 = R_1(V_o/1.25 - 1), \text{ with the convention } R_1 = 240 \Omega$$

The 4-mark answer template (all four PYQs are this, verbatim)

- Draw Fig. 6 $\rightarrow$ state the formula with the I_{ADJ} term, then neglect it (say why: μA against mA) $\rightarrow$ compute R_2 at BOTH ends of the range $\rightarrow$ implement as **fixed resistor + potentiometer** $\rightarrow$ close with the dropout constraint $V_i \geq V_{o(\max)} + 3$ V.

★ All four PYQ designs computed ($R_1 = 240 \Omega$ throughout)

PYQ	Range	R_2 at $V_{\min}$	R_2 at $V_{\max}$	Implementation	Supply needed
2083 Q10 [4]	3–15 V	$240(3/1.25 - 1) = 240 \times 1.4 = \mathbf{336 \Omega}$	$240(15/1.25 - 1) = 240 \times 11 = \mathbf{2640 \Omega}$	336 Ω fixed (≈ 330) + 2304 Ω pot (≈ 2.5 k std)	$V_i \geq 18$ V
2082 Bh Q10 [4]	10–25 V	$240(8 - 1) = \mathbf{1680 \Omega}$	$240(20 - 1) = \mathbf{4560 \Omega}$	1.68 k Ω fixed (≈ 1.8 k) + 2880 Ω pot (≈ 3 k)	$V_i \geq 28$ V
2082 Ba Q12(b) [4]	5–15 V	$240(4 - 1) = \mathbf{720 \Omega}$	$240(12 - 1) = \mathbf{2640 \Omega}$	720 Ω fixed + 1920 Ω pot (≈ 2 k)	$V_i \geq 18$ V
2081 Q12 [4]	10–20 V	$240(8 - 1) = \mathbf{1680 \Omega}$	$240(16 - 1) = \mathbf{3600 \Omega}$	1.68 k Ω fixed + 1920 Ω pot (≈ 2 k)	$V_i \geq 23$ V

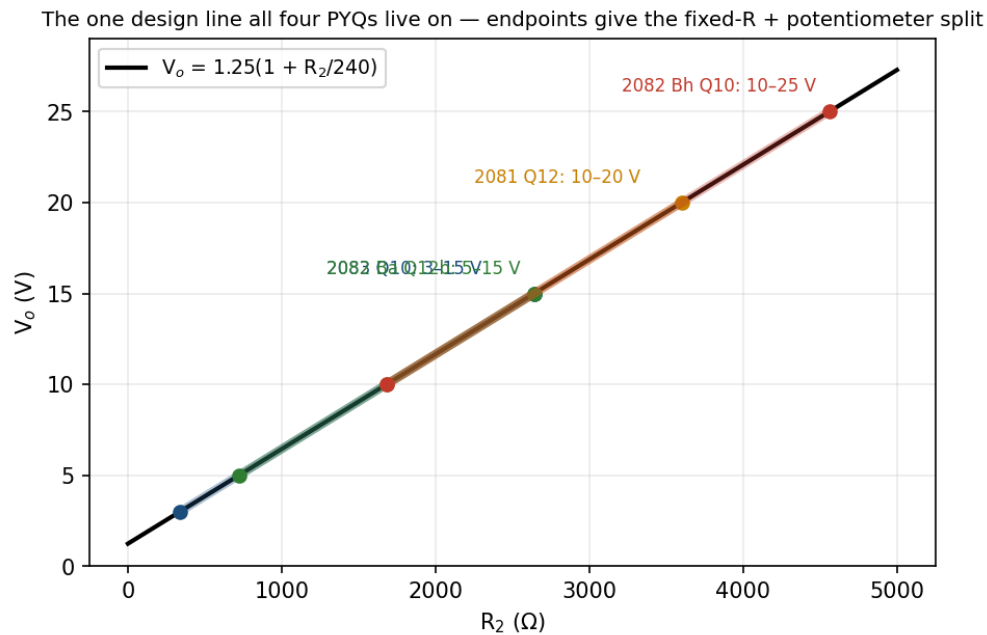


Fig. 7 — All four PYQ designs live on the single line $V_o = 1.25(1 + R_2/240)$, computed and plotted; each range's endpoints give the fixed-plus-pot split. [computed from the design equation]

Model 4-mark answer, fully written (2083 Q10: 3-15 V) — adapt for any range

- **1.** Circuit as Fig. 6; $V_o = 1.25(1 + R_2/R_1) + I_{ADJ}R_2$; $I_{ADJ} \approx 50 \mu A$ is negligible against $I_1 = 1.25/240 \approx 5.2 \text{ mA}$, so $V_o \approx 1.25(1 + R_2/240)$. Choose the standard $R_1 = 240 \Omega$.
- **2.** $V_o = 3 \text{ V} \Rightarrow R_2 = 240(3/1.25 - 1) = \mathbf{336 \Omega}$. $V_o = 15 \text{ V} \Rightarrow R_2 = 240(15/1.25 - 1) = \mathbf{2640 \Omega}$.
- **3.** Realise R_2 as a 336Ω fixed resistor (nearest standard 330Ω) in series with a 2304Ω potentiometer (use $2.5 \text{ k}\Omega$): turning the pot sweeps V_o continuously over 3-15 V.
- **4.** Constraint: the unregulated input must stay $\geq V_{o(max)} + 3 \text{ V dropout} \Rightarrow V_i \geq 18 \text{ V}$. (Bonus polish: the 1.25 V is the IC's internal *bandgap* reference — §5.3.)
- $\triangle$ Mistakes: quoting ONE R_2 for a range question (both ends + pot are required) · forgetting the I_{ADJ} term exists (state it, then neglect it) · omitting the dropout remark · R_1 and R_2 swapped in the formula.

Chapter 5 Final Revision

Conceptual summary (read this the night before)

Rectifier-plus-filter supplies are unregulated: ripple V_r grows with load and V_{dc} rides the mains ($\%VR = (V_{NL} - V_{FL})/V_{FL}$). A zener behind R_S clamps $V_o = V_Z$ but is current-limited and only as stiff as r_z . Putting an emitter follower between zener and load — the series regulator — gives $V_o = V_Z - V_{BE}$ with $\beta \times$ the current, self-correcting because $V_{BE} = V_Z - V_o$; its line rejection is $S_V = r_z/(R + r_z)$. Sampling V_o with a divider and comparing against the zener in an error amplifier makes $V_o = (V_Z + V_{BE2})(R_1 + R_2)/R_2$ — adjustable — and divides both S_V and R_{out} by $(1 + A\beta)$. A sense resistor plus clamp transistor adds current limiting at $I_{L(max)} \approx 0.7/R_{SC}$, turning the regulator into a safe constant-current source under fault. The LM317 packages it all around a 1.25 V bandgap held between OUT and ADJ: $V_o = 1.25(1 + R_2/R_1) + I_{ADJ}R_2$ — and every year's design question is that one line evaluated at the two ends of the requested range, plus the 3 V dropout remark.

Important diagrams checklist

Figure	Priority	Marked in PYQ as
Fig. 6 — LM317 wiring (R_1 OUT→ADJ, R_2 pot to ground)	● HIGH	all four design PYQs ★
Fig. 3(a) — series (emitter-follower) regulator	● HIGH	2083 Q9, 2082 Bh Q9 ★
Fig. 4 — feedback regulator (sampler + error amp)	● HIGH	2081 Q13
Fig. 5 — current-limit add-on ($R_{SC} + Q_3$)	● HIGH	2082 Bh Q9, 2082 Ba Q12(a)
Fig. 7 — LM317 design chart (all four ranges)	● MEDIUM	self-check for any range
Fig. 2 — zener supply	● MEDIUM	foundation; sub-part risk
Fig. 3(b) — shunt regulator	● MEDIUM	contrast lines
Fig. 1 — unregulated chain + ripple	● LOW	never asked standalone

Formula sheet

$\%VR = (V_{NL} - V_{FL})/V_{FL} \times 100$ · $r = V_r(\text{rms})/V_{dc}$ · zener: $I_S = (V_i - V_Z)/R_S = I_Z + I_L$
 Series reg: $V_o = V_Z - V_{BE}$ · shunt: $V_o = V_Z + V_{BE}$ · $S_V = r_z/(R + r_z)$
 Feedback: $V_o = (V_Z + V_{BE2})(R_1 + R_2)/R_2$ · $S_V, R_{out} \div (1 + A\beta)$ · $I_{L(max)} \approx 0.7/R_{SC}$
 LM317: $V_o = 1.25(1 + R_2/R_1) + I_{ADJ}R_2$ · $R_2 = R_1(V_o/1.25 - 1)$ · $R_1 = 240 \Omega$ ·
 dropout: $V_i \geq V_{o(max)} + 3 V$

Priorities / mistakes / derivation list

- **HIGH** LM317 design template with all four range computations (★4/4) · series-regulator family: operation → S_V derivation → feedback improvement → current limiting (★4/4, rotating).
- **MEDIUM** Zener-supply design rules · shunt contrast · bandgap-to-LM317 link · foldback mention.
- **LOW** Ripple formula details · constant-current diode · 78xx/79xx beyond one line.
- △ Top mistakes: $V_o = V_Z + V_{BE}$ for the series type · S_V derived for one topology while drawing another · one R_2 quoted for a range · I_{ADJ} never mentioned · dropout omitted · R_{SC} outside the load path.
- Results to hard-memorise: **1.25 V · 240 Ω · 3 V dropout · 0.7/R_{SC} · $r_z/(R+r_z)$** .

Complete Chapter 5 PYQ Bank (wording preserved)

Year / Exam	Q	Question (condensed, wording preserved)	Mark s	Type
2083 Baishakh (Back)	9	Derive the expression for stability factor of a discrete-transistor series regulator.	4	Derivation
2083 Baishakh (Back)	10	Design a variable DC voltage regulator using the LM317 to output (3–15) volts.	4	Design
2082 Bhadra (Regular)	9	Explain the working principle of a series transistor voltage regulator, accompanied by short-circuit/overload protection.	5	Theory
2082 Bhadra (Regular)	10	Design a 10–25 V variable DC voltage regulator using LM317 IC.	4	Design
2082 Baishakh (Back)	12	Draw series voltage regulator with current limiting circuit and explain how this protection circuit works. Design a voltage regulator to give output voltage from 5 V to 15 V using LM317.	4+4	Theory + Design
2081 Ashwin (Regular)	12	Design variable DC voltage regulator using LM317 to get (10–20) volts output.	4	Design
2081 Ashwin (Regular)	13	Explain the working principle of a series transistor voltage regulator with improved performance and derive an expression for the voltage stability.	2+2	Theory + Derivation

PYQ → What to Study Mapping

PYQ	Topic	Notes section(s)	Required skill
2083 Q9 [4]	Stability factor	\$5.4 (zener model \$5.3)	define $S_V \rightarrow r_z/(R+r_z)$ derivation → interpretation
2083 Q10 [4]	LM317 3–15 V	\$5.6 table row 1 + model answer	formula → $336/2640 \Omega \rightarrow \text{pot} \rightarrow V_i \geq 18 \text{ V}$
2082 Bh Q9 [5]	Series reg + protection	\$5.4 + \$5.5-B (Figs. 3a, 5)	follower action + R_{SC}/Q_3 clamp + $0.7/R_{SC}$
2082 Bh Q10 [4]	LM317 10–25 V	\$5.6 row 2	$1680/4560 \Omega + \text{pot} + V_i \geq 28 \text{ V}$
2082 Ba Q12(a) [4]	Current limiting	\$5.5-B (Fig. 5)	circuit + normal/fault narrative + $I_{\max}$
2082 Ba Q12(b) [4]	LM317 5–15 V	\$5.6 row 3	$720/2640 \Omega + \text{pot}$
2081 Q12 [4]	LM317 10–20 V	\$5.6 row 4	$1680/3600 \Omega + \text{pot} + V_i \geq 23 \text{ V}$
2081 Q13 [2+2]	Improved regulator + stability	\$5.5-A (Fig. 4)	error-amp loop story + V_o formula + $S_V/(1+A\beta)$

Final Coverage Audit

- ✓ Syllabus 5.1–5.6 all present in order; 5.1–5.3 covered at foundation depth despite no standalone PYQ (coverage rule) — and \$5.3's bandgap is cited inside the \$5.6 answer as the LM317's 1.25 V source.
- ✓ References honoured: Boylestad/Floyd (regulator + protection circuits, LM317 pattern with the $R_1 = 240 \Omega$ convention), Millman (stability treatment), Sedra-Smith (zener r_z model, bandgap). Conventions flagged: S_V symbol; the series-vs-shunt V_{BE} sign difference.
- ✓ All 7 PYQs (9 sub-parts) embedded in their sections AND repeated in the bank + mapping; all four LM317 ranges computed with arithmetic shown and plotted on the computed design chart; the cross-topic Q12 split into (a)/(b) and mapped separately.
- ✓ ★ Repeated topics flagged (LM317 4/4, series-regulator family 4/4); ● priorities from PYQ evidence; 7 figures rendered locally — filter ripple and the design chart computed from the governing equations, per the fetch report (0/7 from web).

End of Chapter 5 — and of the series: Chapters 2-5 as full-depth PDFs plus Chapter 1 (BJT) in markdown cover the complete ENEX 151 / EX 151 syllabus with every PYQ from all four papers embedded, solved, and mapped.